

Real-Time ASL Letter Recognition via Landmark-Based Deep Learning

Group members:

(1) Nicholas Offei (nao60)

(2) Rija Khan (rk1047)

(3) Jacob Sze (jms1200)

(4) Piyaporn Puangprasert (pp712)

Course: Introduction to Artificial Intelligence 16:198:520 (SP2026)

Instructor: Professor Casimir A. Kulikowski

Report due date: May 12, 2026

1. Project Goals and Objective

The purpose and goal of this project is to train and evaluate a classifier for accurate sign letter detection. Here, we developed a real time American Sign Language (ASL) model capable of detecting letters from a live webcam feed. The model was then deployed on a website to provide immediate feedback to users seeking to learn sign language for communication.

2. Review/Prior Work and Inspiration

The computer vision foundation of sign language recognition has demonstrated that geometry-based representations of the hand outperform raw pixel-based approaches in robustness and generalization (Bytez.com et al., 2025; Moryossef et al., 2021; Vogler & Metaxas, 1997). Motivated by this principle, we use MediaPipe Hand Landmarker to extract 21 three-dimensional keypoints per frame, reducing each image to a compact structural representation. Unlike pixel-based convolutional models, which achieve strong accuracy but remain computationally intensive and sensitive to variation, landmark-based methods decouple gesture representation from low-level image artifacts (Sun et al., 2025). When trained on normalized geometric features, lightweight models can achieve high accuracy (>90-98%) with real-time performance (Joo et al., 2024). However, a persistent limitation across studies is the ambiguity between visually similar letter pairs (e.g., {S, T}, {M, N}, {U, V}), {P, Q}, {D, T}), reflecting intrinsic constraints of static handshape representation (Tse et al., 2017). Accordingly, we develop a compact, landmark-based 1D convolutional neural network for real-time classification, targeting accuracy comparable to or exceeding heavier pixel-based models while maintaining the computational efficiency required for practical deployment.

3. Preliminary Proposed Project

This problem can be further understood through the lens of intelligent systems. A sign recognition model must transform raw visual input into structured representations and produce reliable decisions under real-time constraints. This Perception → Representation → Decision pipeline provides a principled framework for system design and motivates the use of compact, geometry-driven features over high-dimensional pixel inputs. Within this framework, we focus specifically on real time alphabet-level classification for 24 of the 26 ASL letters (excluding J and Z due to their dynamic motion), as reliable static handshape recognition forms the essential foundation for scalable word- and phrase-level systems.

4. Development of software implementation process

4.1 Dataset Background and Analysis (Source and Composition)

We trained and evaluated on the ASL Dataset hosted on Kaggle (Londhe), which comprises 400× 400 pixel JPEG images of hand signs spanning the full English alphabet (A–Z). Consistent with our focus on static handshapes, we restrict the working corpus to 24 letter classes, excluding J and Z whose canonical ASL forms require dynamic motion trajectories. Each class contains approximately 6000 images, with natural intra-class variation in hand pose, finger extension angle, position in the image, and wrist orientation. The images serve only as inputs to a landmark extractor, not as direct training data for the classifier.

4.2 Preprocessing: Landmark Extraction via MediaPipe

Each image was processed through Google's MediaPipe Hand Landmarker (Zhang et al., 2020) configured for single-hand detection with relaxed confidence thresholds (`min_hand_detection_confidence = 0.1`, `min_tracking_confidence = 0.2`) to maximize

yield from the dataset's tightly cropped frames. The landmarker returns 21 three-dimensional keypoints per detected hand, each corresponding to a specific anatomical joint.

A small portion of the dataset is used due to the sheer amount of images present in the dataset. This effectively reduces time in setting up the data for training and testing. This might not be the case with a more powerful device that can do these computations faster.

Successfully processed samples are represented as (21, 3) tensors of normalized landmark coordinates, where x and y are image-normalized to [0, 1] and z encodes estimated depth relative to the wrist. The resulting dataset spans approximately 5000 samples across 24 classes. The landmarks go through an additional layer of preprocessing where they are marked relative to the wrist, allowing the wrist to act as an origin. We used the following formula to do this:

$$landmark_{i,new} = landmark_i - palm, \text{ where palm is the 0-th landmark}$$

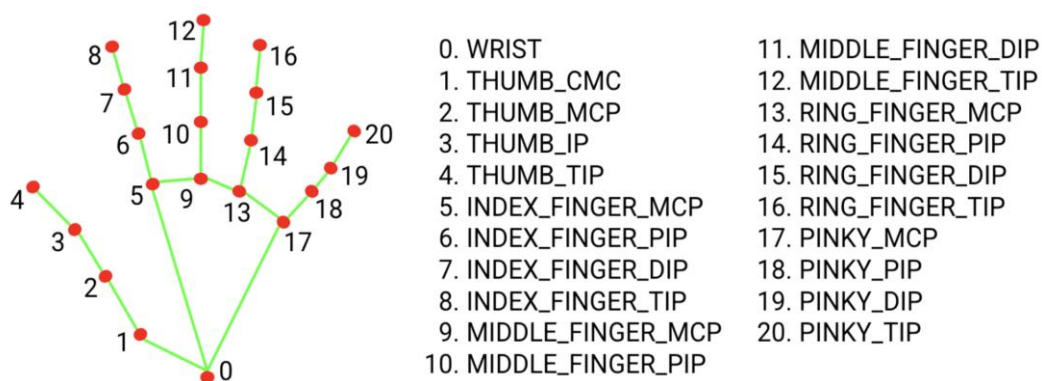


Figure 1: The 21-hand landmark model by MediaPipe
(<https://mediapipe.readthedocs.io/en/latest/solutions/hands.html>)

4.3 Data Partitioning and Evaluation Protocol

The processed landmark dataset was partitioned into training (60%), validation (10%), and test (30%) splits using stratified random sampling, preserving per-class proportions across all three sets. Stratification is not merely a procedural choice here: because MediaPipe detection failures are class-dependent, naive random splitting risks producing test sets that over- or under-represent difficult classes, yielding accuracy estimates that do not faithfully characterize model capability.

All model selection decisions of architecture, regularization, and early stopping criterion were made exclusively on validation performance. The held-out test set was evaluated once, after training was complete, to provide an unbiased estimate of generalization. This discipline is especially important given the small dataset size, where repeated evaluation against a fixed test set could introduce implicit selection bias comparable in magnitude to the generalization gap itself.

5. Software Implementation and Technical Challenges

5.1 Motivation for a 1D CNN on Landmark Sequences

We treat the extracted hand landmarks as a structured spatial sequence rather than a flat feature vector. MediaPipe returns 21 keypoints in a fixed anatomical order such that adjacent entries in the sequence correspond to physically adjacent joints on the hand. A Convolutional Neural Network (CNN) (1-Dimensional CNN) operating over this sequence can learn about local geometric relationships, such as the curl or extension of neighboring joints, directly from that structure.

The alternative (flattening the (21, 3) tensor into a 63-dimensional vector and using a fully-connected network) is simpler but discards the anatomical ordering of the landmarks. The 1D CNN preserves that structure as an inductive bias, which we expect to aid generalization on a small dataset.

5.2 Model Architecture

The input tensor of shape (21, 3) is transposed to (3, 21) to match PyTorch's Conv1d (Lugaresi et al., n.d.; Paszke et al., 2017) convention. Two convolutional blocks are applied in sequence: the first uses 32 filters of kernel size 3 with ReLU and MaxPool1d(2), reducing the sequence from 21 to 9; the second uses 128 filters of kernel size 3 with ReLU and MaxPool1d(2), yielding a feature map of shape (128, 3). This is flattened to 384 dimensions, passed through a 128-unit fully connected layer with ReLU, and projected to 24 output logits, one per target letter class.

5.3 Model Training and Selection

The model was trained with cross-entropy loss and Adam, using mini-batches of size 64. We saved a checkpoint whenever validation loss improved and used the best checkpoint for all final evaluation, which mitigates overfitting without requiring a fixed early stopping rule. Training and validation loss were monitored across epochs to confirm convergence, with validation accuracy tracked alongside as a secondary signal.

5.4 Support Vector Machine (SVM) Baseline

Alongside the CNN, we trained a Support Vector Machine as a diagnostic baseline, not as an alternative deployment candidate, but to identify where the landmark representation itself is fundamentally ambiguous, independent of model capacity. The SVM used an RBF kernel on the flattened 63-dimensional landmark vectors with a one-vs-one multi-class

strategy ($C = 1$, $\gamma = 0.1$). The SVM confusion matrix showed systematic misclassification of T as S, consistent with prior work on confusable ASL pairs (Tse et al., 2017). Inspection of the two classes' landmark scatter plots confirmed that T and S overlap substantially in normalized coordinate space, with thumb placement as the primary but subtle distinguishing feature. This motivates future work on inter-landmark angle encoding and differential depth weighting, as uniform treatment of x, y, and z coordinates does not sufficiently resolve the ambiguity.

5.5 Testing with Real-Time Webcam

The trained model was deployed in a real-time webcam application using OpenCV for video capture and MediaPipe for live landmark extraction. On each frame, the pipeline mirrors the training preprocessing exactly: the frame is flipped horizontally (to produce a natural mirror-image perspective for the user), converted to RGB, passed through the MediaPipe Hand Landmarker in IMAGE mode, the landmarks are normalized to be centered around the wrist (acting as an origin), and the resulting 21 landmarks are extracted into a (1, 21, 3) tensor. The CNN then produced class logits, which are converted to probabilities via softmax; the argmax is displayed on-screen alongside the confidence score.

Critically, the landmark extraction parameters in the webcam script exactly match those used during training (same model asset, same confidence thresholds, same single-hand configuration), ensuring that the distribution of inputs seen at inference time is consistent with the distribution used for training, a key requirement for valid generalization.

6. Experimental Results

The confusion matrix in **Figure 2.A** shows how well the model can classify ASL letters during testing. Certain rows like row 8, which represents the letter I, show the model accurately classifying that letter for most of the time, however it sometimes classify it as the letter S (column 17). This is due to the specific hand positionings the dataset has for these two letters, where the thumb is in front of the fingers. During the training and validation phase, the model learned the features of the hand landmarks very well within the 40 epochs it ran. The plots shown in **Figures 2.B** and **2.C** show how quickly it takes for the model to learn patterns from the hand landmarks, shooting to approximately 85% in validation accuracy in just 5 epochs. The model achieves a test accuracy of around 99.23%, which shows that the model performs well with the dataset used. The classification report for the model is shown in **Figure 3**.

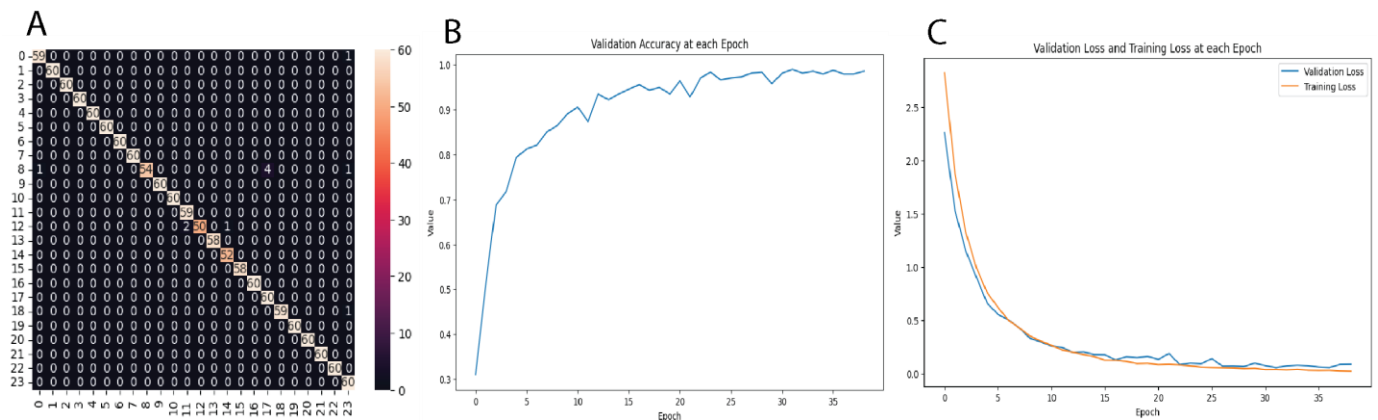


Figure 2. Performance evaluation of the landmark-based 1D CNN model for ASL letter classification. (A) Confusion matrix showing classification performance across the 24 ASL letter classes. (B) Validation accuracy across training epochs, reaching stable convergence around epoch 35. (C) Training and validation loss curves across epochs showing model convergence and reduced overfitting.

Test Accuracy with model.pth: 99.2254%				
	precision	recall	f1-score	support
0	0.9833	0.9833	0.9833	60
1	1.0000	1.0000	1.0000	60
2	1.0000	1.0000	1.0000	60
3	1.0000	1.0000	1.0000	60
4	1.0000	1.0000	1.0000	60
5	1.0000	1.0000	1.0000	60
6	1.0000	1.0000	1.0000	60
7	1.0000	1.0000	1.0000	60
8	1.0000	0.9000	0.9474	60
9	1.0000	1.0000	1.0000	60
10	1.0000	1.0000	1.0000	60
11	0.9672	1.0000	0.9833	59
12	1.0000	0.9434	0.9709	53
13	1.0000	1.0000	1.0000	58
14	0.9811	1.0000	0.9905	52
15	1.0000	1.0000	1.0000	58
16	1.0000	1.0000	1.0000	60
17	0.9375	1.0000	0.9677	60
18	1.0000	0.9833	0.9916	60
19	1.0000	1.0000	1.0000	60
20	1.0000	1.0000	1.0000	60
21	1.0000	1.0000	1.0000	60
22	1.0000	1.0000	1.0000	60
23	0.9524	1.0000	0.9756	60
accuracy			0.9923	1420
macro avg	0.9926	0.9921	0.9921	1420
weighted avg	0.9926	0.9923	0.9922	1420

Figure 3: Classification report of the model after testing.

7. Conclusion: Effectiveness and AI Limitations

Overall, while our model achieved a high-test accuracy of 99.2254%, several limitations would still need further investigation. For example, the current model focuses only on static letter classification, and future work could extend this toward full word recognition. Second, the dataset primarily assumes right-hand usage, so future data augmentation could incorporate left-hand signs as additional classes to make the model comprehensive (**Figure 4 & 5**). Finally, models such as LSTMs or RNNs could be explored for finger spelling and word formation, as they can store sequential context from previous signs.

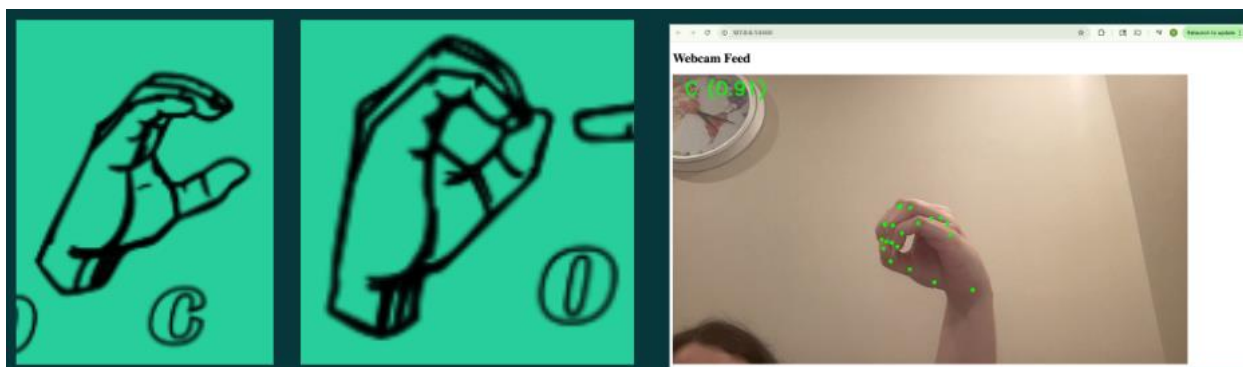


Figure 4: Misclassification of visually similar signs between “O” and “C”. *Example figure showing a classification error. This picture, the user post “O” letter sign, but the application return “C” letter with a confidence score at 0.91, the model is likely misinterpreting of the similar letter sign*

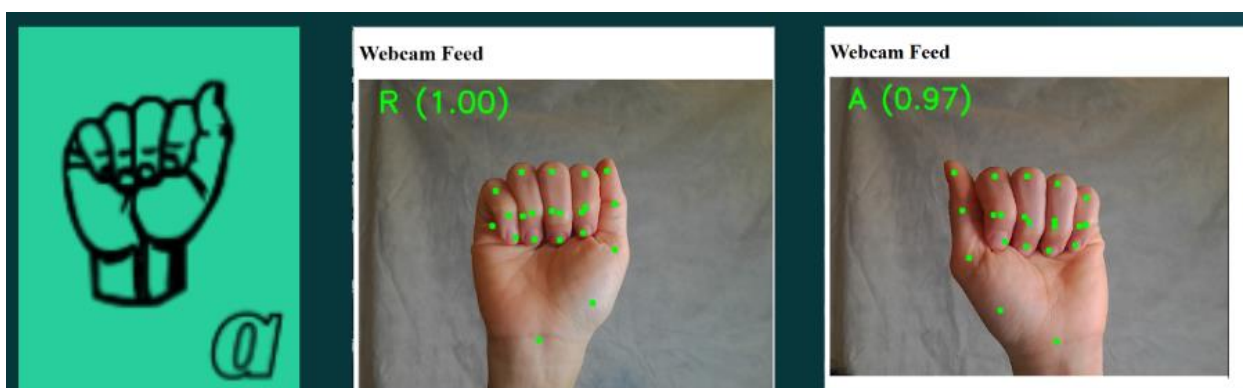


Figure 5: Experiment on left and right hand. *This model trained on the right hand only; we experiment with what the error will present. The result shows most errors will return “R” letter mostly (including front-back palm hand position experiment).*

8. Impact/ Significance of the Applied Problem

Since its formalization in the mid 20th century, American Sign Language (ASL) has served as the primary language of communication for an estimated 500,000 to 2 million deaf and hard-of-hearing individuals in the United States (Mitchell et al., 2006). Yet, to majority of hearing people, ASL remains inaccessible, making a barrier in areas such as healthcare,

education and employment. Employing real-time recognition of ASL gestures bridges this gap and enables fluid communication without a human interpreter. In this project, we have developed a compact, landmark-based 1D convolutional neural network for real-time classification, targeting accuracy comparable to or exceeding heavier pixel-based models while maintaining the computational efficiency required for practical deployment.

The system operates in real time on a standard CPU (no GPU required), confirming that the landmark-based approach achieves the computational efficiency needed for practical deployment, unlike pixel-based ResNet approaches which require GPU acceleration for real-time inference (Chaturvedi, 2025).

9. Reflection and Lessons Learned

Nicholas Offei (nao60): Working on the ASL project helped me understand concepts in feature extraction, pattern recognition, and classification in solving real-world problems. By showing how hand landmarks can be used into structured representations that a model can learn and classify in real time, The project also reinforced how model performance depends not only on the algorithm itself, but also on preprocessing choices and how real-world can be variable.

Rija Khan (rk1047): Overall, this project reinforced the idea that while choosing the right model architecture is important for any project, the interpretation of the data/dataset chosen and its preprocessing is critical for a model's performance. Earlier in the project, we used a dataset that was limited and restrictive in its features, which caused problems during hand landmark extraction for preprocessing and real-time testing of our model. Using this new dataset, which was more diverse and less restrictive in its features, allowed us to reach better performance. Using simpler and smaller size processed data

that can be extracted from the image reduces the model's training time and achieves better results as compared to just using the raw data.

Jacob Sze (jms1200): It was interesting to see what will happen if one uses a complex model sequentially to another complex model. By using the output of MediaPipe, it is able to get significantly better results in a real-world structure. Instead of the basic CNN which would struggle against context the image, MediaPipe is able to capture that using the 21 landmarks. As such, it became quite cool to see how this result really worked out.

Piyaporn Puangprasert (pp712): I learned how the MediaPipe hand landmark works and learned how each of the ASL letter posts. I also learned how to work in a team. For the implementation of the model, the hand position, lighting, and skin tones color have affected the accuracy of the letter model.

10. Group Reflection and Lessons Learned

As a group, we learned both technically and collaboratively throughout this project. We gained experience in machine learning model development, landmark preprocessing, real-time deployment, and evaluating AI systems under real-world conditions. We also learned the importance of communication, coordination, and adaptability, as real-world implementation does not always follow the original proposal, and unexpected challenges can arise during development. Through division of work and supporting one another when needed, we were able to complement each other's strengths and successfully complete the project as a team.

REFERENCES:

Link to Github Repository: <https://github.com/rjj6khan/asl-letter-detection>

- Bytez.com, ABBAHADDU, Y., & Aboussalah, A. M. (2025, December 4). *A Geometry-Aware Metric for Mode Collapse in Time Series...*
<https://bytez.com/docs/neurips/117444/paper>
- Chaturvedi, R. (2025). *Practical GPU Choices for Earth Observation: ResNet-50 Training Throughput on Integrated, Laptop, and Cloud Accelerators* (Version 1). arXiv. <https://doi.org/10.48550/ARXIV.2508.18206>
- Joo, J., Koh, J., & Lee, H. (2024). Hand Gesture Recognition Using Ultrasonic Array with Machine Learning. *Sensors*, 24(20), 6763. <https://doi.org/10.3390/s24206763>
- Kapil Londhe. (2021). American Sign Language [Data set]. Kaggle. <https://doi.org/10.34740/KAGGLE/DSV/2184214>
- Lugaresi, C., Tang, J., Nash, H., McClanahan, C., Uboweja, E., Hays, M., Zhang, F., Chang, C.-L., Yong, M. G., Lee, J., Chang, W.-T., Hua, W., Georg, M., & Grundmann, M. (n.d.). *MediaPipe: A Framework for Perceiving and Processing Reality*.
- Mitchell, R. E., Young, T. A., Bachleda, B., & Karchmer, M. A. (2006). How Many People Use ASL in the United States? Why Estimates Need Updating. *Sign Language Studies*, 6(3), 306–335. <https://doi.org/10.1353/sls.2006.0019>
- Moryossef, A., Tsochantaridis, I., Dinn, J., Camgoz, N. C., Bowden, R., Jiang, T., Rios, A., Muller, M., & Ebling, S. (2021). Evaluating the Immediate Applicability of Pose Estimation for Sign Language Recognition. *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 3429–3435. <https://doi.org/10.1109/CVPRW53098.2021.00382>
- Paszke, A., Gross, S., Chintala, S., Chanan, G., Yang, E., DeVito, Z., Lin, Z., Desmaison, A., Antiga, L., & Lerer, A. (2017). *Automatic differentiation in PyTorch*. <https://openreview.net/forum?id=BJJsrnfCZ>
- Sun, Y., Liu, Z., Zhong, Y., He, H., Yin, H., & Lin, G. (2025). MediaPipe Landmark-Based Inter-Frame Gesture Stability Determination Algorithm. *2025 IEEE 7th International Conference on Communications, Information System and Computer Engineering (CISCE)*, 844–849. <https://doi.org/10.1109/CISCE65916.2025.11065549>
- Tse, R., Li, A., & Chui, Z. (2017). *Detection and Recognition of Sign Language Protocol using Motion Sensing Device*.
- Vogler, C., & Metaxas, D. (1997). Adapting hidden Markov models for ASL recognition by using three-dimensional computer vision methods. *Computational Cybernetics and Simulation 1997 IEEE International Conference on Systems, Man, and Cybernetics*, 1, 156–161 vol.1. <https://doi.org/10.1109/ICSMC.1997.625741>
- Zhang, F., Bazarevsky, V., Vakunov, A., Tkachenka, A., Sung, G., Chang, C.-L., & Grundmann, M. (2020). *MediaPipe Hands: On-device Real-time Hand Tracking* (arXiv:2006.10214). arXiv. <https://doi.org/10.48550/arXiv.2006.10214>